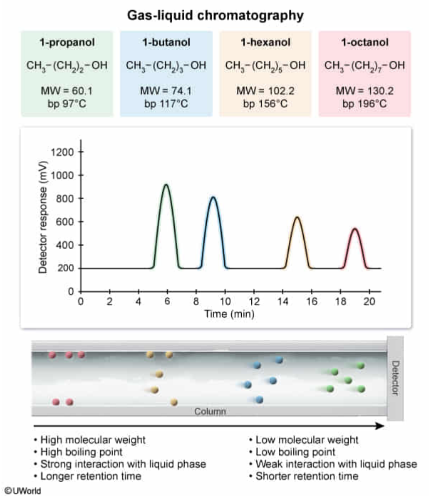
A mixture of 1-butanol, 1-hexanol, 1-octanol, and 1-propanol is separated by gas-liquid chromatography. What compound elutes last?

- A. 1-butanol [4%]
- B. 1-hexanol [1%]
- C. 1-octanol [70%]
- D. 1-propanol [24%]

Correct
70% answered correctly

Explanation:



Gas-liquid chromatography (GC) is a technique used to separate molecules in a mixture, primarily on the basis of **boiling point**. The mixture to be separated is injected into a chamber that is heated to a temperature above the boiling points of all compounds in the mixture. All the compounds vaporize and enter the column, which is coated with a thin liquid layer that acts as the stationary phase. The mobile phase is an inert gas with a low boiling point (eg, N_2, He, or Ar).

Compounds in the mixture that have higher boiling points more readily reenter the liquid (stationary) phase, and therefore move more slowly through the column and elute later.

Relative boiling points of molecules can be determined based on intermolecular forces, **molecular weight** (MW), and branching. The molecules in the mixture, 1-propanol, 1-butanol, 1-hexanol, and 1-octanol, are all alcohols with three-, four-, six-, and eight-carbon chains, respectively. The chain length can be determined by the compound's prefix.

Because the molecules in the question all have the same functional group (alcohol) but different carbon counts, their relative boiling points depend on MW only. A **higher MW** indicates a **higher boiling point** and **longer retention time** because molecules with larger MWs have larger surface areas, giving a greater extent of London forces between the molecules. **1-Octanol** has the greatest number of carbons (eight), the highest MW, and the highest boiling point of the four alcohols in the mixture and thus corresponds to the **last peak** in the GC trace (ie, has the longest retention time).

(Choices A, B, and D) 1-Butanol, 1-hexanol, and 1-propanol have smaller carbon chains, lower MWs, and consequently lower boiling points than 1-octanol. These three alcohols therefore elute before 1-octanol.

Educational objective:
Gas-liquid chromatography separates molecules based on boiling point. A number of factors contribute to a molecule's boiling point: intermolecular forces, molecular weight, and branching. For compounds with the same functional group but different carbon counts, a higher molecular weight indicates a higher boiling point and a longer retention time.

Time spent: 0 sec
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Organic Chemistry

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